

LAB TASK 2

Question 1):

Instruction	Value of AX	Value of BX	Value of CX
Mov ax, 5	0005 (5)	0000 (0)	0012 (18)
Mov bx ,10	0005 (5)	000A (10)	0012 (18)
add ax, bx	000F (15)	000A (10)	0012 (18)
Mov cx, 15	000F (15)	000A (10)	000F (15)
add cx, bx	000F (15)	000A (10)	0019 (25)

Explanation:

Mov ax, 5:

- ✓ When this line of code runs.
- ✓ AX register holds **0005h (5 decimal)**.
- ✓ The value of registers BX and CX will be the garbage values (I run the program 2 times. First time I got the value of CX as 0012, and second time I got the value 0000).
- ✓ Value of BX register is 0000.
- ✓ Value of CX register is 0012 (18 in decimal).

Mov bx, 10:

- ✓ AX register holds the same value i.e. 0005.
- ✓ BX register holds **000A (10)**.
- ✓ CX register value remains the same whatever provided by DosBox. In my case it is (0012).

Add ax, bx:

- ✓ This command means
 - **Add** the values of registers **ax + bx** and store the result in register **AX**.
 - AX register now holds **000F (15)**.
- ✓ Bx register holds the same **000A (10)**.
- ✓ CX register's story is same (0012).

Mov cx, 15:

- ✓ AX register holds 000F (15).
- ✓ BX register holds (000A) (10).
- ✓ CX register will now holds value **000F (15)**.

Add cx, bx:

- ✓ AX register = 000F (15).
- ✓ Bx register = 000A (10).

- ✓ CX register will now holds **0019 (25 in decimal)**.
 - This command means add the values of bx and cx and add the value into Cx register.

Question 2:

Program:

```
[org 0x0100]
```

```
mov ax, 7
mov bx, 10
add ax, bx
mov cx, 20
add cx, ax
mov ax, 0x4c00
int 0x21
```

Output:

[illegible]

- ✓ Mov ax, 7 (AX holds 0007).
- ✓ Mov bx, 10 (BX now holds value 000A).

- ✓ Add ax, bx ($ax = ax + bx = 0007 + 000A$)
- ✓ $AX = 0011$
- ✓ Cx holds value of 0014 (20).

AX	0011	SI	0000	CS	19F5	IP	0108	Stack	+0	0000	Flags	7214
DX	0004	DI	0000	DS	19F5				+2	20C0		
CX	001A	BP	0000	ES	19F5	HS	19F5		+4	9FFF	OF	DF
SI	0000	SP	FFFF	SS	19F5	FS	19F5		+6	EAno	0	0
											1	0
											1	0
CHD VS or SI or SP 0 1 2 3 4 5 6 7												
0108	B91409	MU	AX,0014									
0108	01C1	ADD	CX,AX									
010D	B8004C	MU	AX,4000									
0110	CD21	INT	21									
0112	EB04	JMP	0118									
0114	31D2	XDR	DX,DX									
0116	31C8	XDR	AX,AX									
0118	B956E4	MU	[BP-1C],DX									
011B	B946E6	MU	[BP-1A],AX									
DS:0000 C0 20 FF 9F 00 EA 0F FE AD DE 1B 05 C5 06 00 00 00 00 DS:0010 18 01 10 01 18 01 92 01 01 01 0F 00 01 0F 00 01 0F 00 DS:0020 FF FF FF FF FF FF FF FF FF FF 1B 00 11 00 00 00 00 DS:0030 A2 01 14 00 18 00 F5 19 FF FF FF 00 00 00 00 00 00 DS:0040 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 = f_0 = 11 11 11 11 11 11 11 11 11 11 11 11 11 11 11 11 11												
1	Step	2	ProcStep	3	Retrieve	4	Help ON	5	BRK Menu	6	7	up
8	dn	9	ie	10	ri							

```

AX 0011 SI 0000 CS 19F5 IP 0100 Stack +0 0000 Flags 7200
BX 000A DI 0000 DS 19F5 +2 20CD
CX 0025 BP 0000 ES 19F5 HS 19F5 +4 9FFF OF DF IF SF ZF AF PF CF
DX 0000 SP FFFE SS 19F5 FS 19F5 +6 EA00 0 0 1 0 0 0 0 0

3 or SI or SY
CMD >S

010B 01C1 ADD CX,AX DS:0000 CD 20 FF 9F 00 EA F0 FE
010D B8004C MOV AX,4C00 DS:0000 AD DE 1B 05 C5 06 00 00
0110 CD21 INT 21 DS:0010 18 01 10 01 18 01 92 01
0112 EB04 JMP 0110 DS:0010 01 01 01 00 FF 00 01 FF
0114 31D2 XOR DX,DX DS:0020 FF FF FF FF FF FF FF FF
0116 31C0 XOR AX,AX DS:0020 FF FF FF FF EB 19 C0 11
0118 8956E4 MOV [BP-1C1],DX DS:0030 A2 01 14 00 18 00 F5 19
011B 8946E6 MOV [BP-1A1],AX DS:0030 FF FF FF FF 00 00 00 00
011E C746F60000 MOV [BP-0A1],0000 DS:0040 05 00 00 00 00 00 00 00

2 0 1 2 3 4 5 6 7 8 9 A B C D E F
DS:0000 CD 20 FF 9F 00 EA F0 FE AD DE 1B 05 C5 06 00 00 = f.a= i|.t...
DS:0010 18 01 10 01 18 01 92 01 01 01 01 00 FF 00 01 FF .....f. ....
DS:0020 FF FF FF FF FF FF FF FF FF FF FF FF EB 19 C0 11 .....δ.L.
DS:0030 A2 01 14 00 18 00 F5 19 FF FF FF FF 00 00 00 00 6.....J. ....
DS:0040 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

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```

- ✓ Add cx,ax.
- ✓ This command means add cx + ax and store the value in cx register.
- ✓ Cx holds 0025 (37).

Question 3:

Program:

[org 0x0100]

```

mov ax, 70
mov bx, 41
add ax, bx
mov dx, 85
add dx, ax
mov ax, 0x4c00
int 0x21

```

Output:

- ✓ AX = 0046 (70)
- ✓ BX = 0029 (41)
- ✓ CX = 0012 (random value by DOSBOX).
- ✓ DX = 0000 (random value by dosbox).

```

AX 0046 SI 0000 CS 19F5 IP 0106 Stack +0 0000 Flags 7200
BX 0029 DI 0000 DS 19F5 +2 20CD
CX 0012 BP 0000 ES 19F5 HS 19F5 +4 9FFF OF DF IF SF ZF AF PF CF
DX 0000 SP FFFE SS 19F5 FS 19F5 +6 EA00 0 0 1 0 0 0 0 0

3 or SI or SY
CMD >S

0103 B82900 MOV BX,0029 DS:0000 CD 20 FF 9F 00 EA F0 FE
0106 01D8 ADD AX,BX DS:0000 AD DE 1B 05 C5 06 00 00
0108 BA5500 MOV DX,0055 DS:0010 18 01 10 01 18 01 92 01
010B 01C2 ADD DX,AX DS:0010 01 01 01 00 02 FF FF FF
010D B8004C MOV AX,4C00 DS:0020 FF FF FF FF FF FF FF FF
0110 CD21 INT 21 DS:0020 FF FF FF FF EB 19 C0 11
0112 EB04 JMP 0110 DS:0030 A2 01 14 00 18 00 F5 19
0114 31D2 XOR DX,DX DS:0030 FF FF FF FF 00 00 00 00
0116 31C0 XOR AX,AX DS:0040 05 00 00 00 00 00 00 00

2 0 1 2 3 4 5 6 7 8 9 A B C D E F
DS:0000 CD 20 FF 9F 00 EA F0 FE AD DE 1B 05 C5 06 00 00 = f.a= i|.t...
DS:0010 18 01 10 01 18 01 92 01 01 01 01 00 02 FF FF FF .....f. ....
DS:0020 FF FF FF FF FF FF FF FF FF FF FF FF EB 19 C0 11 .....δ.L.
DS:0030 A2 01 14 00 18 00 F5 19 FF FF FF FF 00 00 00 00 6.....J. ....
DS:0040 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

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```

AX 006F	SI 0000	CS 19F5	IP 0100	Stack +0 0000	Flags 7204
BX 0029	DI 0000	DS 19F5		+2 20CD	
CX 0012	BP 0000	ES 19F5	HS 19F5	+4 9FFF	OF DF IF SF ZF AF PF CF
DX 0055	SP FFFE	SS 19F5	FS 19F5	+6 EA00	0 0 1 0 0 1 0 0
S or SI or SP					
CMD >S				0 1 2 3 4 5 6 7	
0100 BA500	MOV	DX,0055	DS:0000	CD 20 FF 9F 00 EA F0 FE	
0100 01C2	ADD	DX,AX	DS:0000	AD DE 1B 05 C5 06 00 00	
0100 B804C	MOV	AX,4C00	DS:0010	18 01 10 01 18 01 92 01	
0110 CD21	INT	21	DS:0020	FF FF FF FF FF FF FF FF	
0112 EB04	JMP	0118	DS:0028	FF FF FF FF EB 19 C0 11	
0114 31D2	XOR	DX,DX	DS:0030	A2 01 14 00 18 00 F5 19	
0116 31C0	XOR	AX,AX	DS:0038	FF FF FF FF 00 00 00 00	
0118 8956E4	MOV	[BP-1C],DX	DS:0040	05 00 00 00 00 00 00 00	
011B 8946E6	MOV	[BP-1A],AX	DS:0048	00 00 00 00 00 00 00 00	
2 0 1 2 3 4 5 6 7 8 9 A B C D E F					
DS:0000	CD 20 FF 9F 00 EA F0 FE	AD DE 1B 05 C5 06 00 00	= f.0= i.+....		
DS:0010	18 01 10 01 18 01 92 01	01 01 01 00 02 FF FF FF	f.		
DS:0020	FF FF FF FF FF FF FF FF	FF FF FF FF EB 19 C0 11			
DS:0030	A2 01 14 00 18 00 F5 19	FF FF FF FF 00 00 00 00	0.....J.		
DS:0040	05 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00			

- ✓ AX = 006F (by adding ax and bx).
- ✓ BX = 0029(same as before).
- ✓ CX = 0012 (same as before).
- ✓ DX = 0055 (85).

- ✓ AX = 006F (by adding ax and bx).
- ✓ BX = 0029(same as before).
- ✓ CX = 0012 (same as before).
- ✓ DX = 00C4 (dx = dx + ax).

AX	006F	SI	0000	CS	19F5	IP	0100	Stack	+0 0000	Flags	7210						
BX	0029	DI	0000	DS	19F5				+2 20CD								
CX	0012	BP	0000	ES	19F5	HS	19F5		+4 9FFF	OF	DF	IF	SF	ZF	AF	PF	CF
DX	00C4	SP	FFFF	SS	19F5	FS	19F5		+6 EA00	0	0	1	0	0	1	0	0
S or SI or SP																	
CMD >S								DS:0000 0 1 2 3 4 5 6 7									
0100 01C2 ADD DX,AX								DS:0008 AD DE 1B 05 C5 06 00 00									
0100 B804C MOV AX,4C00								DS:0010 18 01 10 01 18 01 92 01									
0110 CD21 INT 21								DS:0018 01 01 01 00 02 FF FF FF									
0112 EB04 JMP 0118								DS:0020 FF FF FF FF FF FF FF									
0114 31D2 XOR DX,DX								DS:0028 FF FF FF FF EB 19 C0 11									
0116 31C0 XOR AX,AX								DS:0030 A2 01 14 00 18 00 F5 19									
0118 8956E4 MOV [BP-1C],DX								DS:0038 FF FF FF FF 00 00 00 00									
011B 8946E6 MOV [BP-1A],AX								DS:0040 05 00 00 00 00 00 00 00									
011E C746F60000 MOV [BP-0A],0000								DS:0048 00 00 00 00 00 00 00 00									
2 0 1 2 3 4 5 6 7 8 9 A B C D E F																	
DS:0000 CD 20 FF 9F 00 EA F0 FE AD DE 1B 05 C5 06 00 00 = f.0= i.+....																	
DS:0010 18 01 19 01 18 01 92 01 01 01 01 00 02 FF FF FFf.																	
DS:0020 FF FF FF FF FF FF FF FF FF FF FF FF EB 19 C0 11																	
DS:0030 A2 01 14 00 18 00 F5 19 FF FF FF FF 00 00 00 00 0.....J.																	
DS:0040 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00																	
1 Step 2ProcStep3Retrieve4Help ON 5BRK Menu 6 7 up 8 dn 9 le 10 r																	

Question 4:

Program:

[org 0x0100]

```

mov ax, 100
mov bx, 8
mov cx, 22
sub ax, bx
sub ax, cx
mov ax, 0x4c00
int 0x21

```

Output:

AX 0064	SI 0000	CS 19F5	IP 0109	Stack +0 0000	Flags 7200									
BX 0008	DI 0000	DS 19F5		+2 20CD										
CX 0016	BP 0000	ES 19F5	HS 19F5	+4 9FFF	OF DF IF SF ZF AF PF CF									
DX 0000	SP FFFE	SS 19F5	FS 19F5	+6 EA00	0 0 1 0 0 0 0 0									
S or SI or SP														
CMD >S				0 1 2 3 4 5 6 7										
0106 B91600	MOV	CX,0016		DS:0000	CD 20 FF 9F 00 EA F0 FE									
0109 29D0	SUB	AX,BX		DS:0008	AD DE 1B 05 C5 06 00 00									
010B 29C8	SUB	AX,CX		DS:0010	18 01 10 01 18 01 92 01									
010D B8004C	MOV	AX,4C00		DS:0018	01 01 01 00 02 FF FF FF									
0110 CD21	INT	21		DS:0020	FF FF FF FF FF FF FF									
0112 EB04	JMP	011B		DS:0028	FF FF FF FF EB 19 C0 11									
0114 31D2	XOR	DX,DX		DS:0030	A2 01 14 00 18 00 F5 19									
0116 31C0	XOR	AX,AX		DS:0038	FF FF FF FF 00 00 00 00									
011B 8956E4	MOV	[BP-1C],DX		DS:0040	05 00 00 00 00 00 00 00									
2						= f.0= i.+.+								
DS:0000	CD 20 FF 9F 00 EA F0 FE	AD DE 1B 05 C5 06 00 00							f.					
DS:0010	18 01 10 01 18 01 92 01	01 01 01 00 02 FF FF FF							d.L					
DS:0020	FF FF FF FF FF FF FF	FF FF FF FF EB 19 C0 11							6.....J.					
DS:0030	A2 01 14 00 18 00 F5 19	FF FF FF FF 00 00 00 00												
DS:0040	05 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00												
1 Step 2ProcStep 3Retrieve 4Help ON 5BRK Menu 6 7 up 8 dn 9 le 10 ri														

- ✓ Ax = 0064 (100)
- ✓ BX = 0008 (8)
- ✓ CX = 0016 (22)
- ✓ DX = 0000

- ✓ AX changed to 005C (subtract bx from ax and store result in ax register).
- ✓ BX = 0008
- ✓ CX = 0016
- ✓ DX = 0000

AX 005C	SI 0000	DS 19F5	IP 010B	Stack +0 0000	Flags 7214
BX 0008	DI 0000	DS 19F5		+2 20CD	
CX 0016	BP 0000	ES 19F5	HS 19F5	+4 9FFF	OF DF IF SF ZF AF PF CF
DX 0000	SP FFFE	SS 19F5	FS 19F5	+6 EA00	0 0 1 0 0 1 1 0
R or SI or SP					
CMD >S				1 0 1 2 3 4 5 6 7	
0109 29D8	SUB	AX,BX		DS:0000	CD 20 FF 9F 00 EA F0 FE
010B 29C8	SUB	AX,CX		DS:0008	AD DE 1B 05 C5 06 00 00
010D B8004C	MOV	AX,4C00		DS:0010	18 01 10 01 18 01 92 01
0110 CD21	INT	21		DS:0018	01 01 01 00 02 FF FF FF
0112 EB04	JMP	011B		DS:0020	FF FF FF FF EB 19 C0 11
0114 31D2	XOR	DX,DX		DS:0028	A2 01 14 00 18 00 F5 19
0116 31C0	XOR	AX,AX		DS:0030	FF FF FF FF 00 00 00 00
011B 8956E4	MOV	[BP-1C],BX		DS:0038	05 00 00 00 00 00 00 00
011B 8946E6	MOV	[BP-1A],AX		DS:0040	00 00 00 00 00 00 00 00
2					
DS:0000	CD 20 FF 9F 00 EA F0 FE	AD DE 1B 05 C5 06 00 00	= f.0= i.+.+		
DS:0010	18 01 10 01 18 01 92 01	01 01 01 00 02 FF FF FF	f.		
DS:0020	FF FF FF FF FF FF FF	FF FF FF FF EB 19 C0 11	d.L		
DS:0030	A2 01 14 00 18 00 F5 19	FF FF FF FF 00 00 00 00	6.....J.		
DS:0040	05 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00			
1 Step 2ProcStep 3Retrieve 4Help ON 5BRK Menu 6 7 up 8 dn 9 le 10 ri					

AX 0046	SI 0000	CS 19F5	IP 010D	Stack +0 0000	Flags 7200
BX 0008	DI 0000	DS 19F5		+2 20CD	
CX 0016	BP 0000	ES 19F5	HS 19F5	+4 9FFF	OF DF IF SF ZF AF PF CF
DX 0000	SP FFFE	SS 19F5	FS 19F5	+6 EA00	0 0 1 0 0 0 0 0
(R) register value					
CMD >SS				0 1 2 3 4 5 6 7	
010B 29C8	SUB	AX,CX		DS:0000	CD 20 FF 9F 00 EA F0 FE
010D B8004C	MOV	AX,4C00		DS:0008	AD DE 1B 05 C5 06 00 00
0110 CD21	INT	21		DS:0010	18 01 10 01 18 01 92 01
0112 EB04	JMP	011B		DS:0018	01 01 01 00 02 FF FF FF
0114 31D2	XOR	DX,DX		DS:0020	FF FF FF FF EB 19 C0 11
0116 31C0	XOR	AX,AX		DS:0028	A2 01 14 00 18 00 F5 19
011B 8956E4	MOV	[BP-1C],DX		DS:0030	FF FF FF FF 00 00 00 00
011B 8946E6	MOV	[BP-1A],AX		DS:0038	05 00 00 00 00 00 00 00
011E C746F60000	MOV	[BP-0A],0000		DS:0040	00 00 00 00 00 00 00 00
2					
DS:0000	CD 20 FF 9F 00 EA F0 FE	AD DE 1B 05 C5 06 00 00	= f.0= i.+.+		
DS:0010	18 01 10 01 18 01 92 01	01 01 01 00 02 FF FF FF	f.		
DS:0020	FF FF FF FF FF FF FF	FF FF FF FF EB 19 C0 11	d.L		
DS:0030	A2 01 14 00 18 00 F5 19	FF FF FF FF 00 00 00 00	6.....J.		
DS:0040	05 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00			
1 Step 2ProcStep 3Retrieve 4Help ON 5BRK Menu 6 7 up 8 dn 9 le 10 ri					

- ✓ AX = 0046 (subtract ax from cx)
- ✓ All other values remains the same.
- ✓ BX = 0008
- ✓ CX = 0016

Question 5:

Program:

[org 0x0100]

```
mov ax, 10
mov bx, 15
add ax, bx
mov cx, 5
sub ax, cx
mov dx, ax
mov ax, 0x4c00
int 0x21
```

Output:

- ✓ AX = 000A (10)
- ✓ BX = 000F (15)
- ✓ CX = 0014 (Random/garbage values).
- ✓ DX = 0000 (Random/garbage values).

AX	000A	SI	0000	CS	19F5	IP	0106	Stack	+0 0000	Flags	7200
BX	000F	DI	0000	DS	19F5				+2 20CD		
CX	0014	BP	0000	ES	19F5	HS	19F5		+4 9FFF	OF	DF
DX	0000	SP	FFFF	SS	19F5	FS	19F5		+6 EA00	IF	SF
0 1 2 3 4 5 6 7											
DS:0000 CD 20 FF 9F 00 EA F0 FE											
DS:0010 18 01 10 01 18 01 92 01											
DS:0020 FF FF FF FF FF FF FF FF											
DS:0030 A2 01 14 00 18 00 F5 19											
DS:0040 05 00 00 00 00 00 00 00											
DS:0048 00 00 00 00 00 00 00 00											

AX	0019	SI	0000	CS	19F5	IP	010B	Stack	+0 0000	Flags	7210
BX	000F	DI	0000	DS	19F5				+2 20CD		
CX	0005	BP	0000	ES	19F5	HS	19F5		+4 9FFF	OF	DF
DX	0000	SP	FFFF	SS	19F5	FS	19F5		+6 EA00	IF	SF
0 1 2 3 4 5 6 7											
DS:0000 CD 20 FF 9F 00 EA F0 FE											
DS:0010 18 01 10 01 18 01 92 01											
DS:0020 FF FF FF FF FF FF FF FF											
DS:0030 A2 01 14 00 18 00 F5 19											
DS:0040 05 00 00 00 00 00 00 00											
DS:0048 00 00 00 00 00 00 00 00											

- ✓ AX = 0019 (25) ; ax = ax + bx
- ✓ BX = 000f (Same).
- ✓ CX = 0005 ; mov cx, 5
- ✓ DX = 0000

- ✓ AX = 0014 (sub ax – cx = 25 -5 =20)
- ✓ BX = 000F
- ✓ CX = 0005
- ✓ DX = 0014 (moving ax to dx).

AX	0014	SI	0000	CS	19F5	IP	010F	Stack	+0 0000	Flags	7204	
BX	000F	DI	0000	DS	19F5				+2 20CD			
CX	0005	BP	0000	ES	19F5	HS	19F5		+4 9FFF	OF DF IF SF ZF AF PF CF		
DX	0014	SP	FFFE	SS	19F5	FS	19F5		+6 EA00	0 0 1 0 0 0 1 0		
S or SI or SP												
CMD >S								1				0 1 2 3 4 5 6 7
010D	89C2	MOV DX,AX						DS:0000	CD 20 FF 9F 00 EA F0 FE			
010F	B8004C	MOV AX,4C00						DS:0008	AD DE 1B 05 C5 06 00 00			
0112	CD21	INT 21						DS:0010	18 01 10 01 18 01 92 01			
0114	31D2	XOR DX,DX						DS:0018	01 01 01 00 02 FF FF FF			
0116	31C0	XOR AX,AX						DS:0020	FF FF FF FF FF FF FF FF			
0118	8956E4	MOV [BP-1C],DX						DS:0028	FF FF FF FF EB 19 C0 11			
011B	8946E6	MOV [BP-1A],AX						DS:0030	A2 01 14 00 18 00 F5 19			
011E	C746F60000	MOV [BP-0A],0000						DS:0038	FF FF FF FF 00 00 00 00			
0123	8B46F6	MOV AX,[BP-0A]						DS:0040	05 00 00 00 00 00 00 00			
2	0	1	2	3	4	5	6	7	8	9	A B C D E F	
DS:0000	CD	20	FF	9F	00	EA	F0	FE	AD	DE	1B 05 C5 06 00 00	= f.0= i .+...
DS:0010	18	01	10	01	18	01	92	01	01	01	01 00 02 FF FF FF	a.
DS:0020	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF EB 19 C0 11	L.
DS:0030	A2	01	14	00	18	00	F5	19	FF	FF	FF FF 00 00 00 00	6.....J.
DS:0040	05	00	00	00	00	00	00	00	00	00	00 00 00 00 00 00	
1	Step	2	ProcStep	3	Retrieve	4	Help ON	5	BRK Menu	6	7 up 8 dn 9 lc 10 ri	